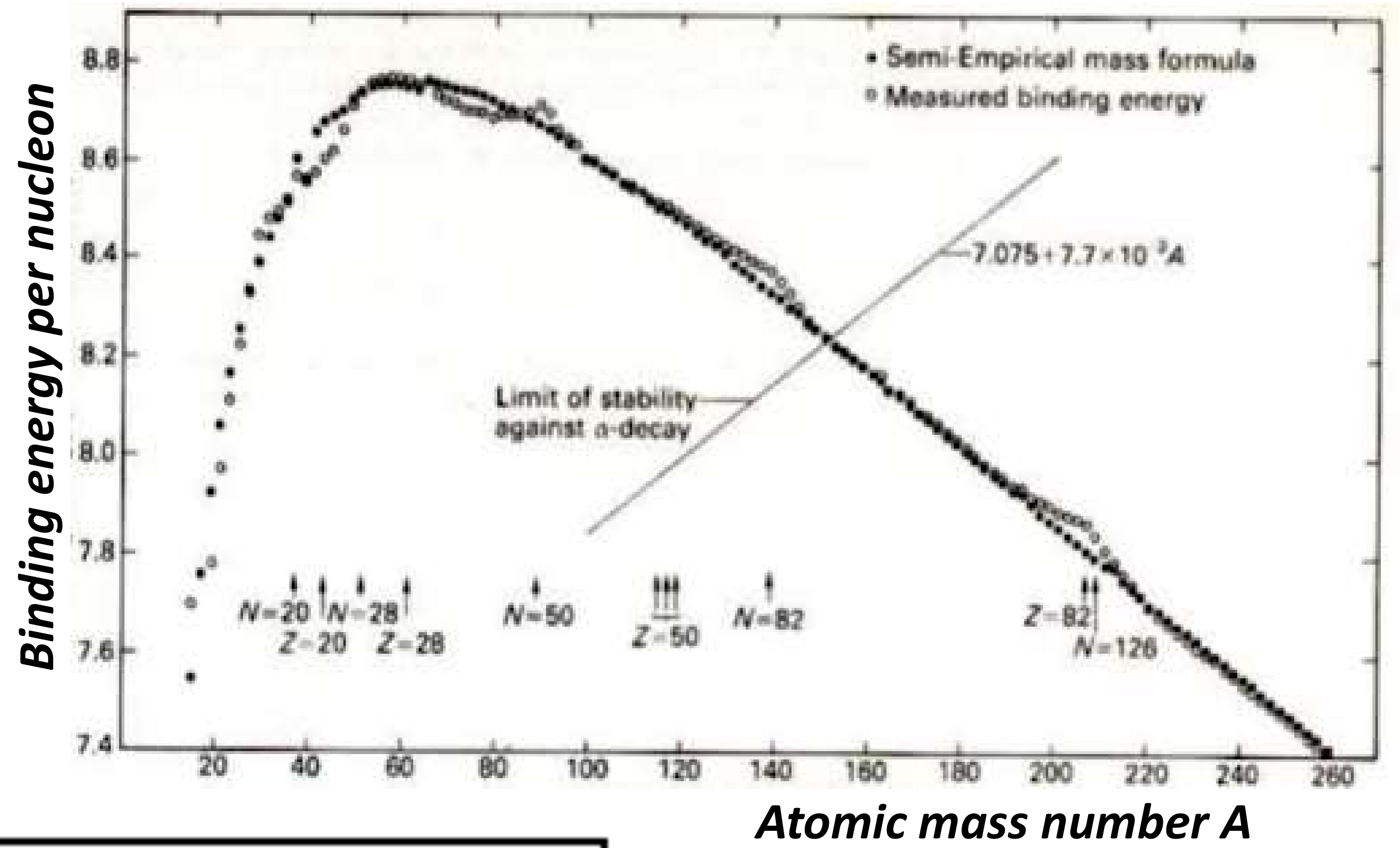
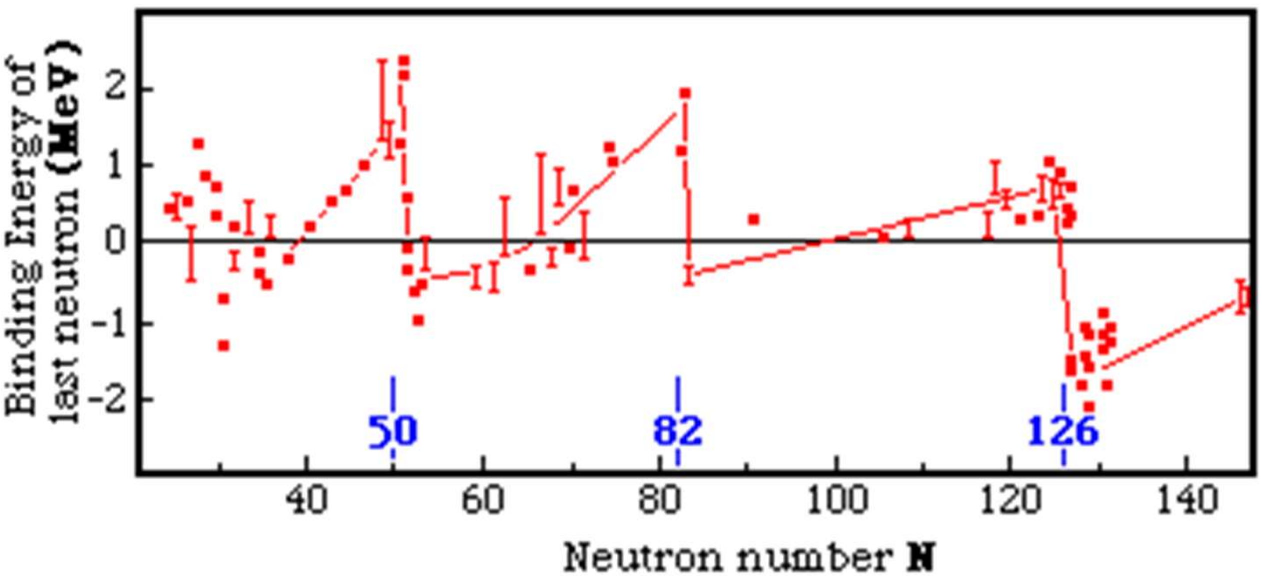
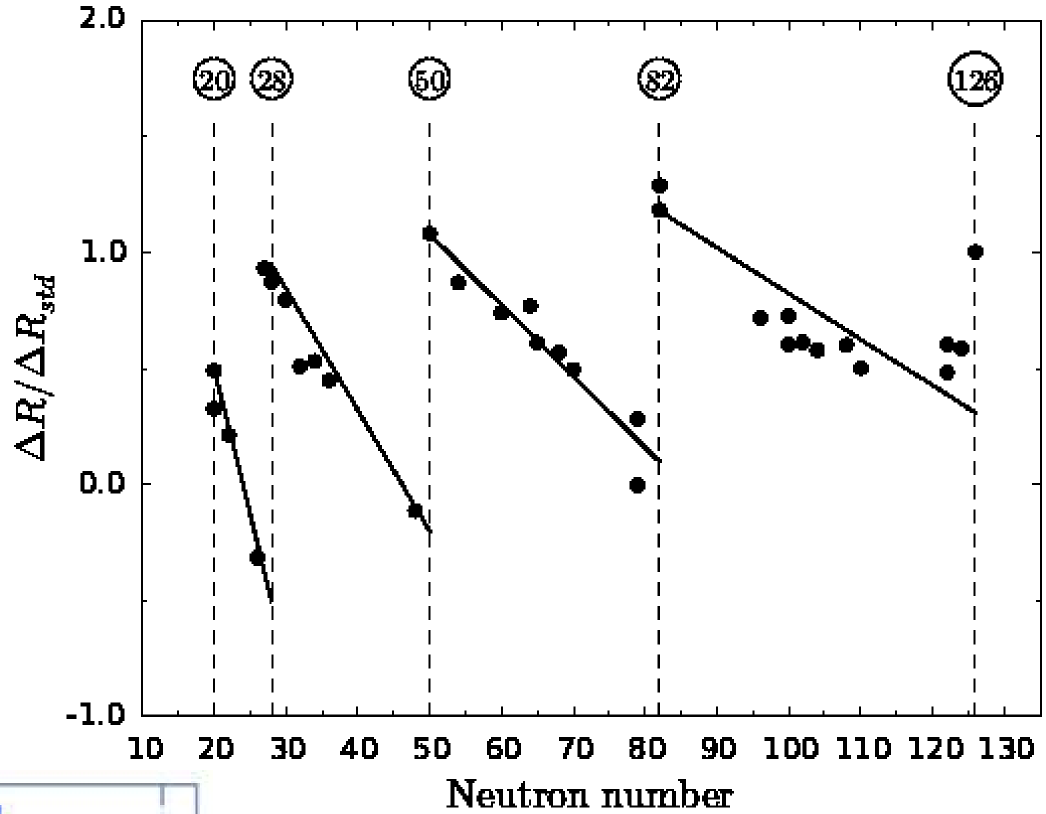
Nuclear Shell Model

Evidence of Nuclear Shell Structure

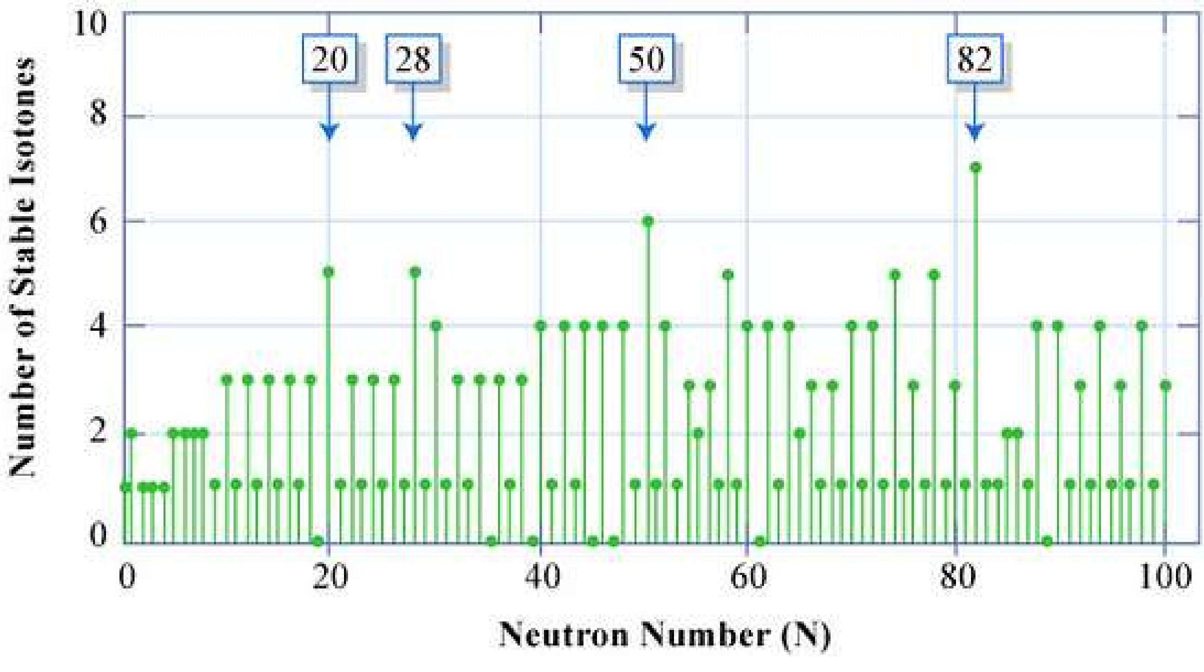




Evidence of Nuclear Shell Structure

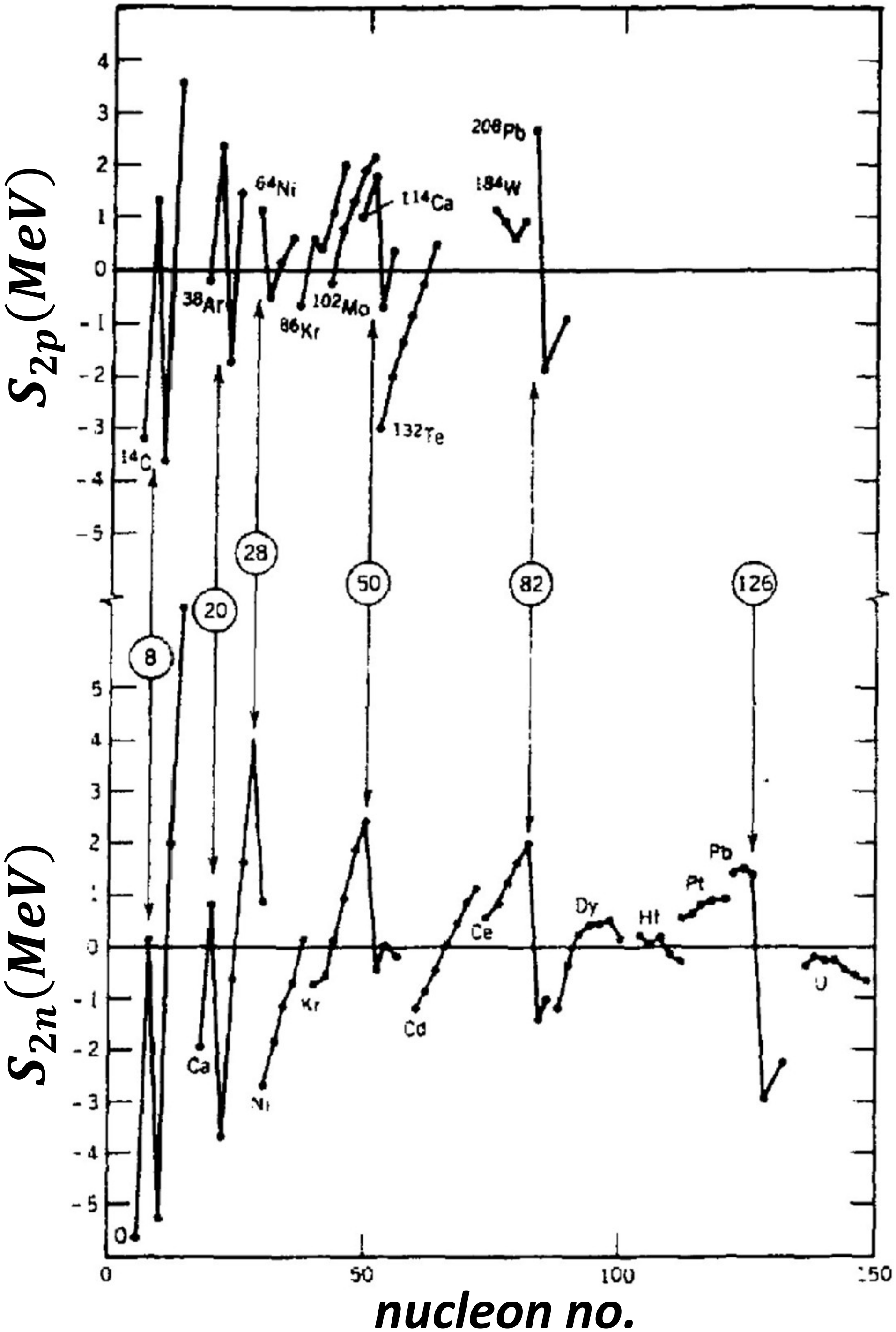


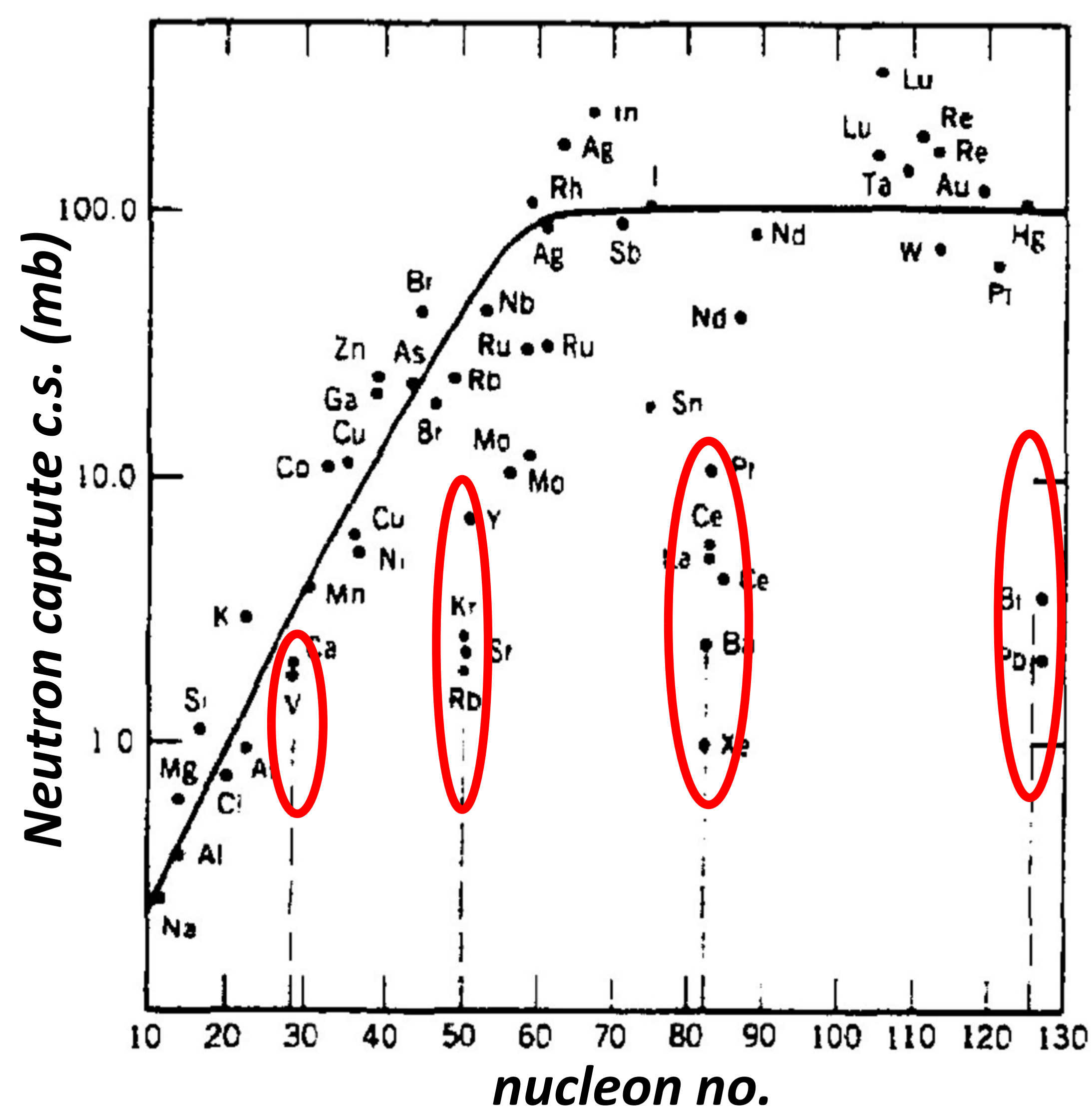
ΔR: *radius change with* $\Delta N = 2$



Evidence of Nuclear Shell Structure

$$S_n = [M(A-1,Z) + M_n - M(A,Z)]c^2$$





Indicates wider spacing of the energy levels just beyond a closed shell

Evidence of Nuclear Shell Structure

- ***The sudden and discontinuous behavior occurs at the same proton and neutron numbers***

- ***All these experimental evidences show***
  - ***some kind of shell structure and some kind of shell closure***
  - **wider spacing of the energy levels just beyond a closed shell**

- ***The nucleon numbers for which the shells close are 2, 8, 20, 28, 50, 82, 126 : Magic numbers***

- ***The magic numbers represent the effects of filled major shells***

- ***Any successful theory must be able to account for this existence of shell closures at those occupation numbers***

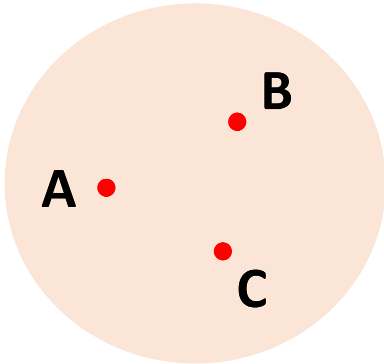
Let's understand

Evidence of Shell Structure

- *Above Deuteron, all nuclei have more than 2 nucleons*

- ***Many-body forces come into picture***

- *For n number of nucleons, there can be n-body interactions*



$$F_A = F_{A,B} + F_{A,C} + F_{A,BC}$$

2-body forces *3-body force*

- ***Even if we know the nucleon-nucleon interaction very well, we cannot write the Hamiltonian for the entire interaction inside the nucleus because of 3-body, 4-body……. Many-body forces also present along with 2-body forces***

- *For a big nucleus, even 2-body interactions for all pairs of nuclei is difficult to write and solve*

- ***Any nucleon in a nucleus is assumed to move in a common potential produced by the force exerted on it by all other nucleons: Mean potential***

Nuclear Potential

$$\hat{H} = \sum_{i=1}^{A} \frac{\hat{p}_i^2}{2m_i} + \sum_{i<j}^{A} \hat{V}(r_i, r_j)$$

$$\hat{H} = \sum_{i=1}^{A} \left[\frac{\hat{p}_i^2}{2m_i} + \hat{V}(r_i) \right]$$

- *We assume a single-particle potential: the field that a nucleon is seeing because of the rest of the nucleons*
- *This we approximate, to the first order, by a central potential*
- *Every nucleon experience the same potential*
- *Whole nucleus gives an average potential to each nucleon: mean potential*

$$\left[-\frac{\hbar^2}{2m} \nabla^2 + V(r) - \varepsilon \right] \Psi(r) = 0$$

$$\Psi(r) = \frac{u_\ell(r)}{r} \cdot Y_{\ell m}(\vartheta, \varphi) \cdot \chi_{m_s}$$

Shell Model Potential

- ***To understand,***
  - ***we assume a central potential***
  - ***solve the Hamiltonian and get energy levels***
  - ***fill the nucleons in these energy levels***
  - ***If we find gaps in-between some of the energy levels, shell structure will come up***

Step 1: Choice of the potential

Shell Model Potential

1. Infinite Square Well Potential

$$V(\vec{r}) = 0 \; for \; r < r_0$$
$$= \infty \; for \; r > r_0$$

$$\varphi(\vec{r}) = \frac{u(r)}{r} Y_l^{m_l}(\theta, \phi)$$

$$l = 0, 1, 2, 3, 4, 5, 6$$
$$s, p, d, f, g, h, i$$

$$-\frac{\hbar^2}{2m}\frac{d^2u}{dr^2} + \left[V(r) + \frac{l(l+1)\hbar^2}{2mr}\right]u = E\,u$$

Solution ($r < r_0$): *(Spr. Bessel's func.)*

$$rj_r(kr) \qquad k = \sqrt{\frac{2mE}{\hbar^2}}$$

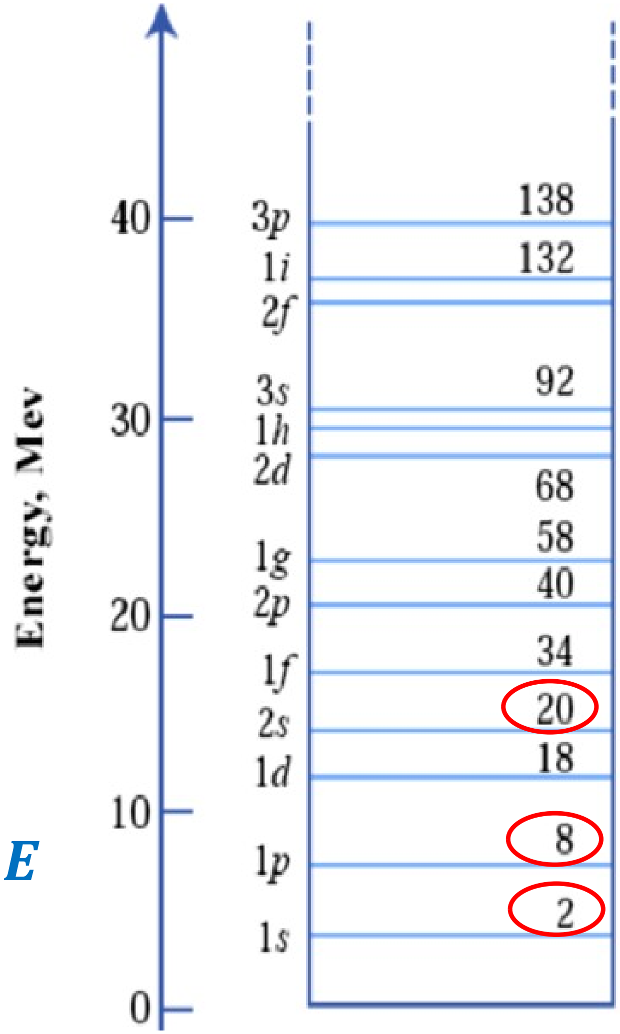
$$rj_r(kr_0) = 0$$

Single-particle energies

$l = 0(s), n = 1, 2, 3 \ldots$ *gives a set of values for* k, E

$l = 1(p), n = 1, 2, 3 \ldots$ *gives another set of values for* k, E

$l = 2(d), n = 1, 2, 3 \ldots$ *gives next set of values for* k, E



Observed magic numbers : 2, 8, 20, 28, 50, 82, 126

Square well potential does not reproduce the observed magic numbers

Shell Model Potential

2. 3-Dim. Harmonic Potential

$$V(\vec{r}) = \frac{1}{2} m\omega^2 r^2 \qquad \varphi(\vec{r}) = \frac{u(r)}{r} Y_l^{m_l}(\theta, \phi)$$

$l = 0, 1, 2, 3, 4, 5, 6$
s, p, d, f, g, h, i

$$-\frac{\hbar^2}{2m}\frac{d^2u}{dr^2} + \left[\frac{1}{2} m\omega^2 r^2 + \frac{l(l+1)\hbar^2}{2mr}\right] u = E\, u$$

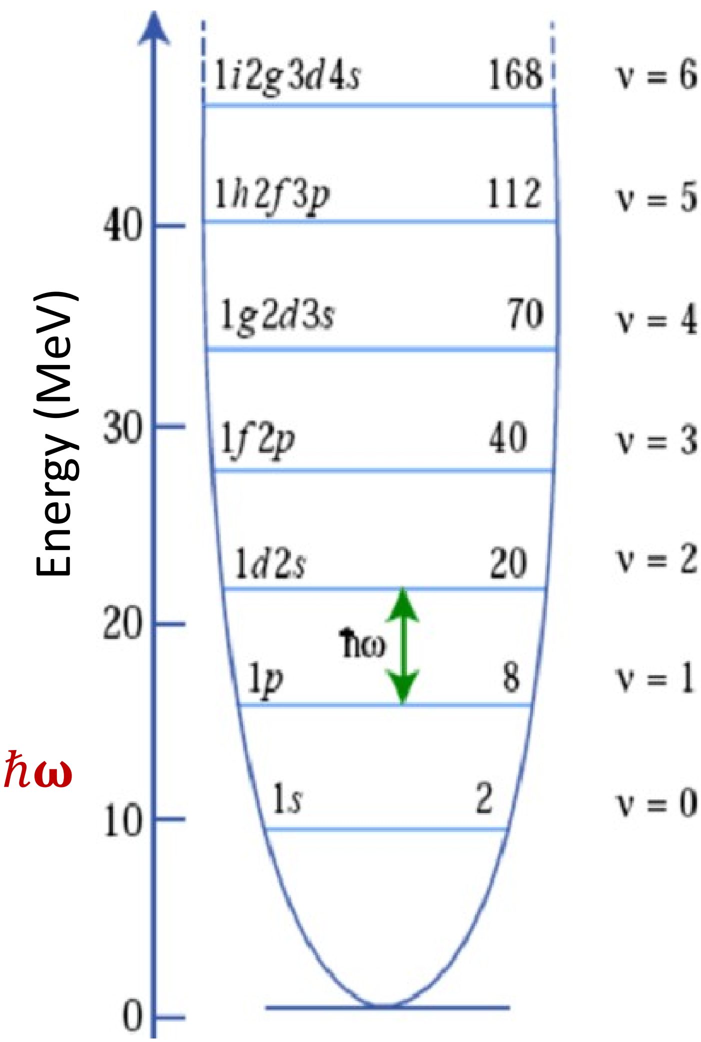
$$E = \left(2n + l + \frac{3}{2}\right)\hbar\omega \qquad n = 0, 1, 2, 3, 4, \ldots \quad l = 0, 1, 2, 3, 4, \ldots$$

$$E = \left(v + \frac{3}{2}\right)\hbar\omega \qquad v = 2n + l$$

Lowest energy $\Rightarrow v = 0\ (\mathrm{l} = 0, \mathrm{n} = 0) \Rightarrow \mathbf{1s} \Rightarrow \mathrm{E} = \frac{3}{2}\hbar\omega$

Next higher energy $\Rightarrow v = 1\ (\mathrm{l} = 1, \mathrm{n} = 0) \Rightarrow \mathbf{1p} \Rightarrow \mathrm{E} = \frac{5}{2}\hbar\omega$

2nd higher energy $\Rightarrow v = 2\ (\mathrm{l} = 2, \mathrm{n} = 0), (\mathrm{l} = 0, \mathrm{n} = 1)$
$\Rightarrow \mathbf{1d, 2s} \Rightarrow \mathrm{E} = \frac{7}{2}\hbar\omega$



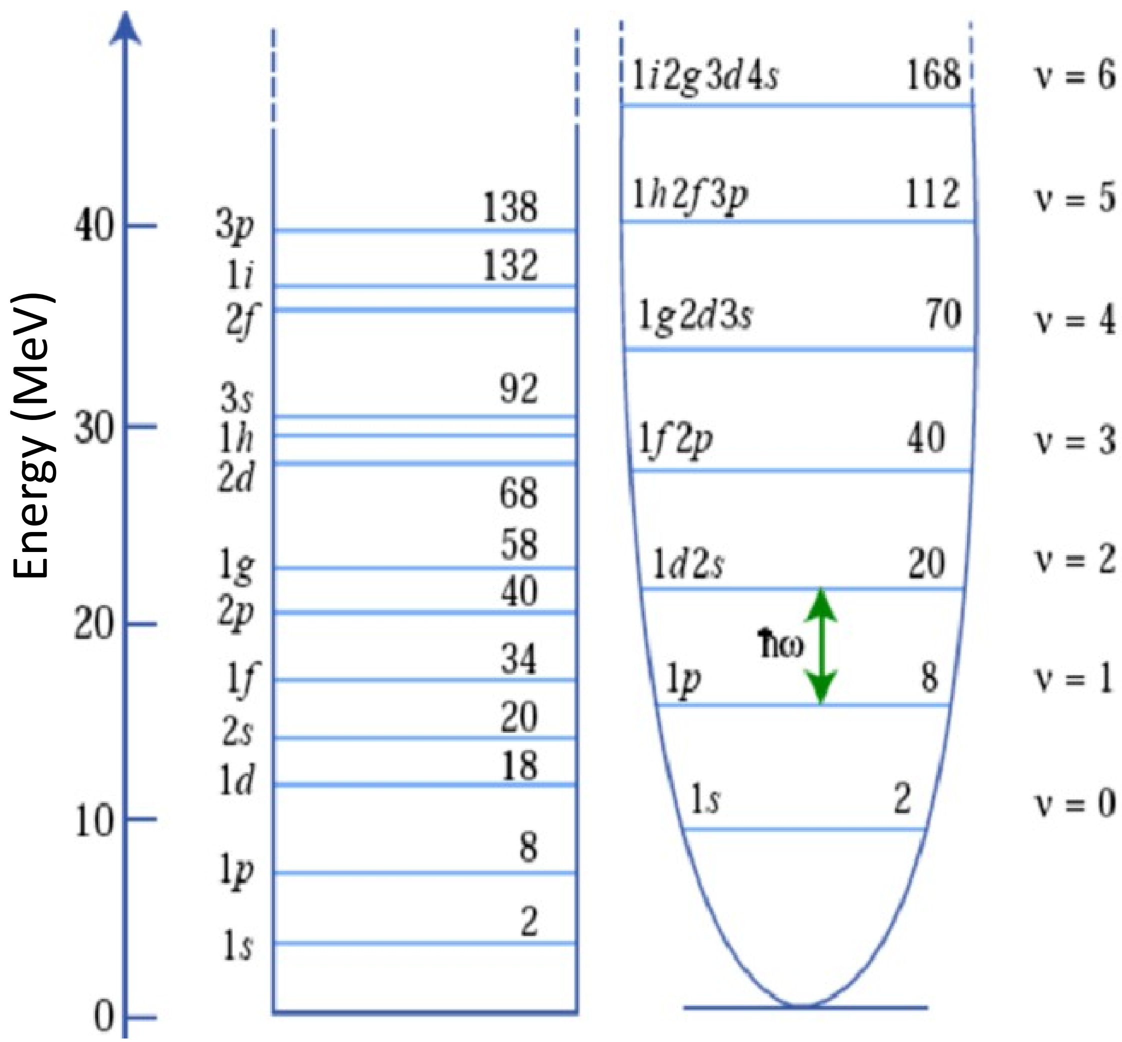
Shell Model Potential

2. Infinite Harmonic Oscillator

$l = 0, 1, 2, 3, 4, 5, 6$
s, p, d, f, g, h, i

- ***Both fail to explain the magic numbers beyond 20***
- ***Both are un-realistic to be considered as nuclear potential***

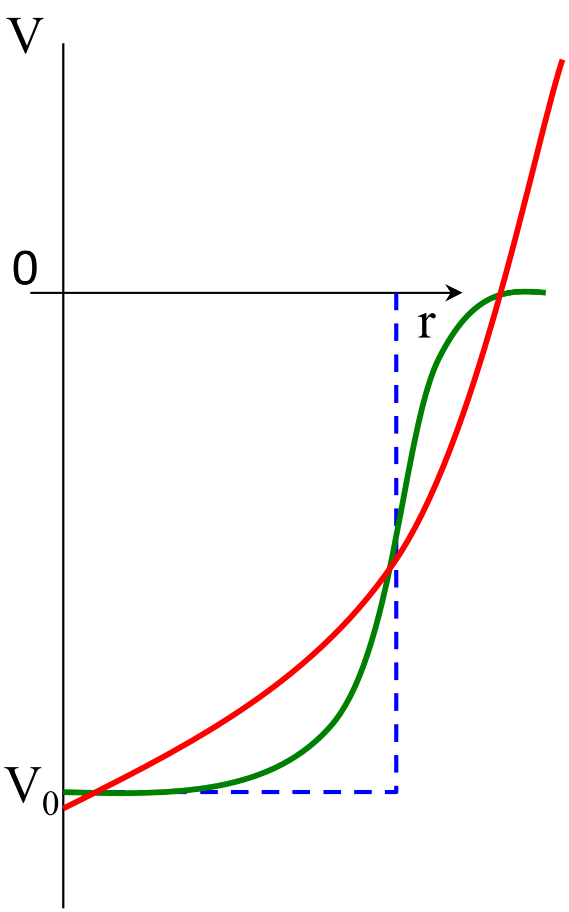
- ***Infinite potentials means infinite amount of energy is needed to take a nucleon out from the nucleus***
- ***But nucleons can be taken out giving in a finite separation energy***
- ***Nuclear potential is finite***
- ***Nuclear potential does not have a sharp edge***
- ***Nuclear potential closely approximates the nuclear charge and matter distribution, falling smoothly to zero beyond the mean radius R***

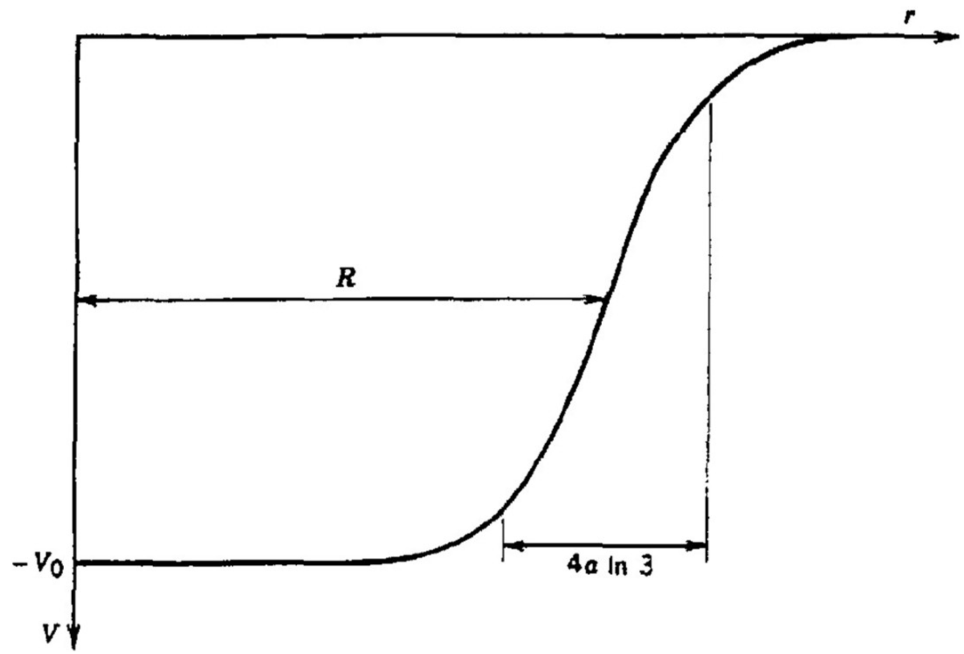


Shell Model Potential

3. Woods-Saxon Potential

$$V(r) = \frac{-V_0}{1 + e^{(r-R/a)}}$$

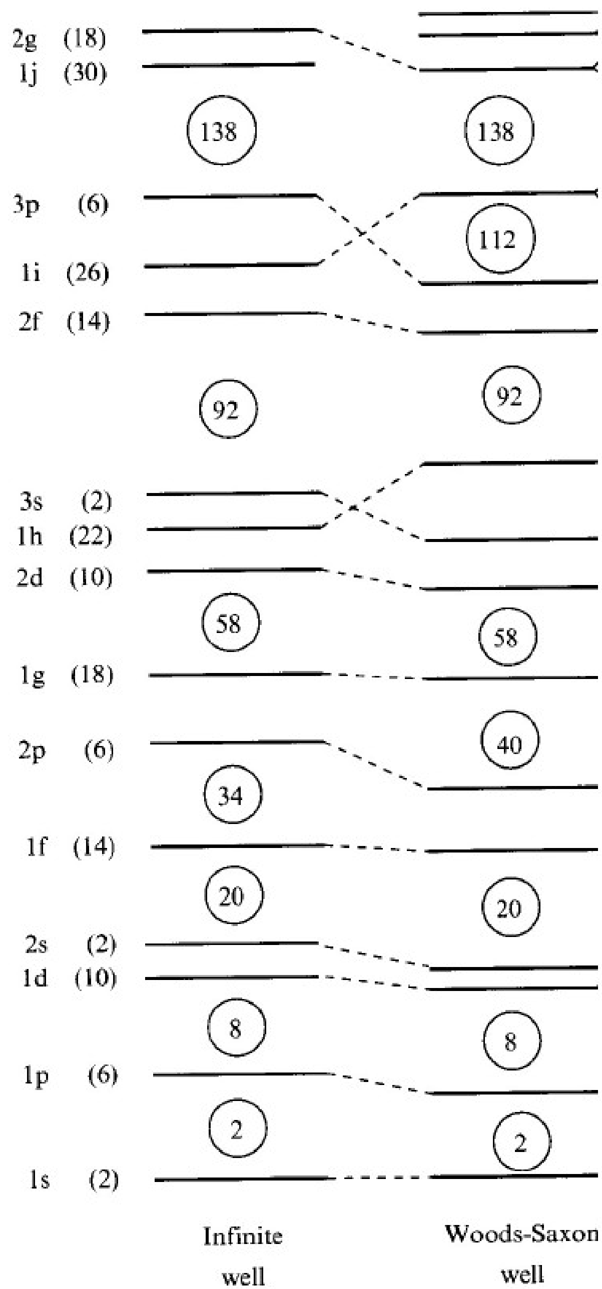




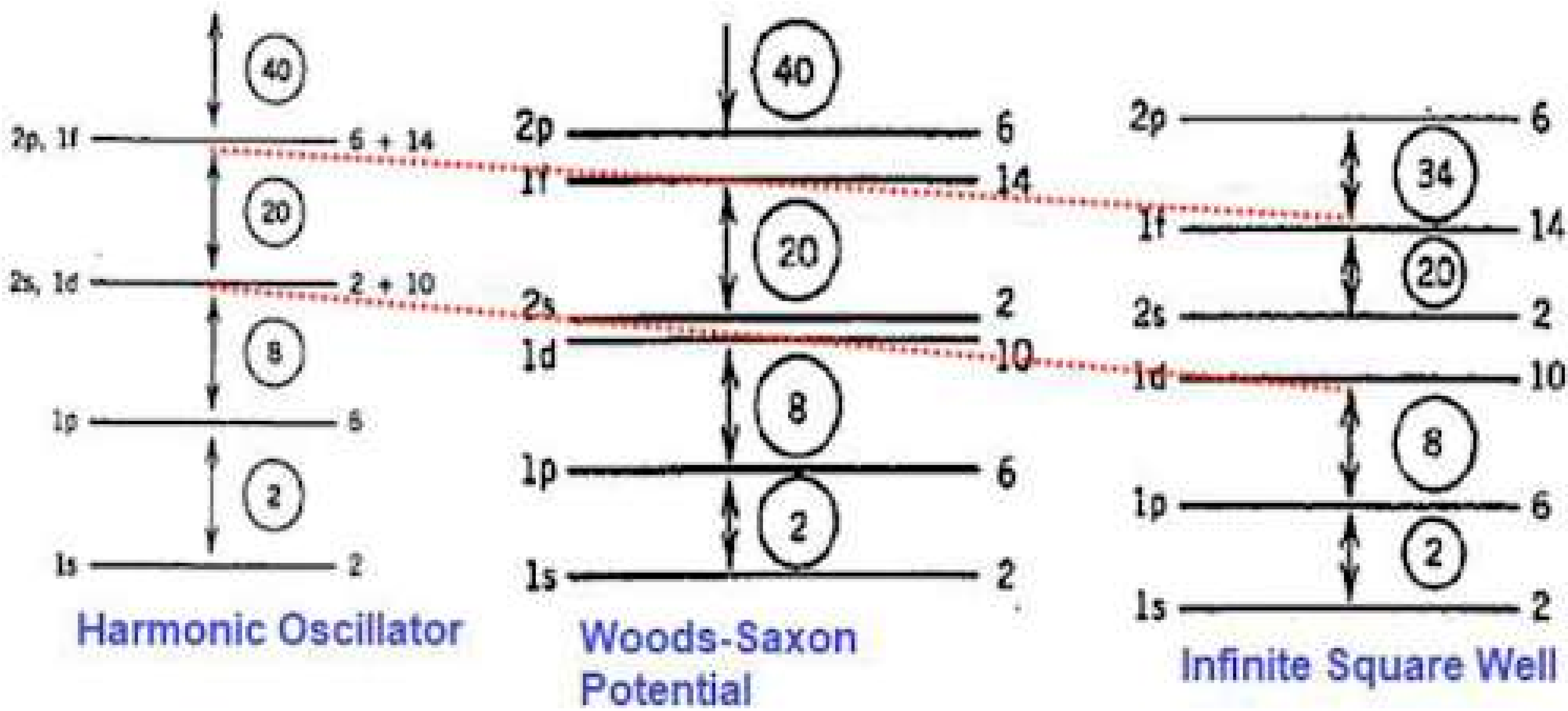
R: mean radius $= 1.25A^{1/3}$
a: skin thickness $= 0.524\ fm$
V_0: well depth $\sim 50\ MeV$
(adjusted to give proper separation energies)

Shell Model Potential

3. Woods-Saxon Potential



Shell Model Potential



$2(2l+1)$ filling of nucleons in each energy level

Magic numbers above 20 do not emerge from the calculations

Shell Model Potential

How can we modify the potential to give the proper magic numbers ?

- ***We cannot make a radical change in the potential because we do not want to destroy the physical content of the model***

- ***The Woods-Saxon potential is already a very good guess at how the nuclear potential should look***

- ***It is therefore necessary to add various terms to this potential to try to improve the situation***

- ***In 1949, Mayer, Haxel, Suess and Jensen showed that the inclusion of spin-orbit potential could give the proper separation of the subshells***

$$H = -\frac{\hbar^2}{2m}\nabla^2 + V(r) + V_{ls}(r)\,\vec{L}\cdot\vec{S}$$

Shell Model Potential

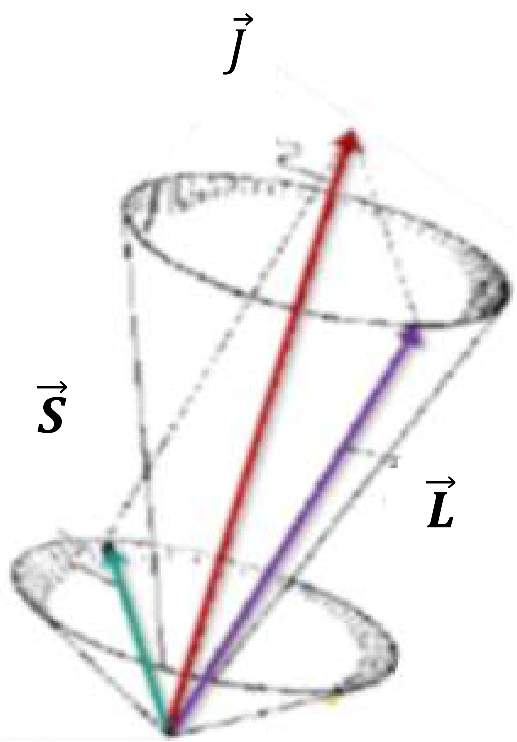
$$\boldsymbol{H = H_0 + V_{ls}(r)\vec{L}\cdot\vec{S}}$$

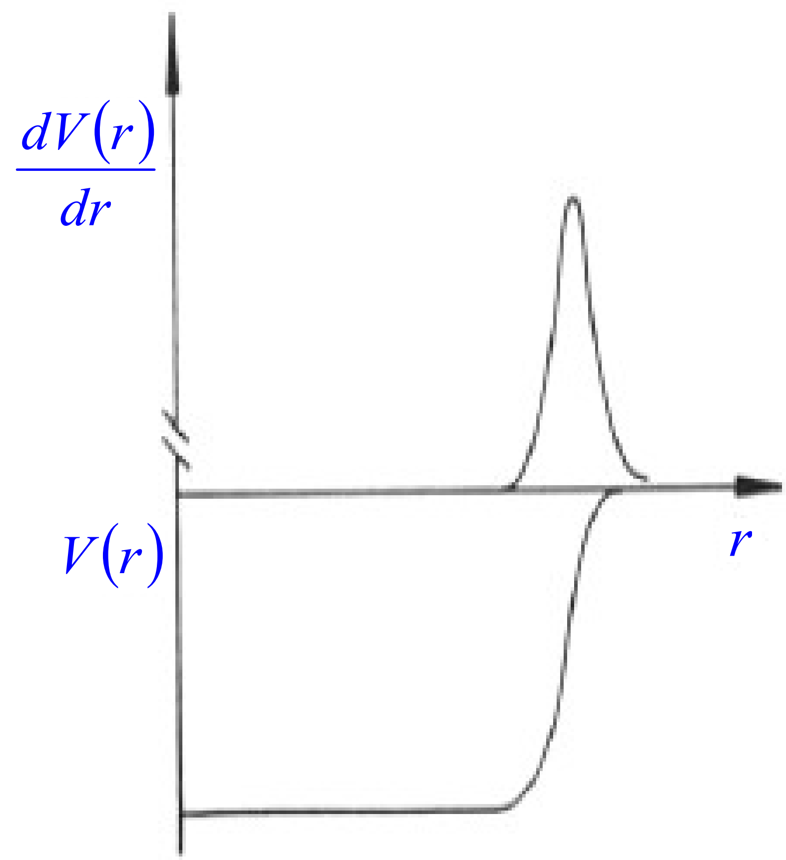
$$\boldsymbol{H_0 = -\frac{\hbar^2}{2m}\nabla^2 + V(r)}$$

1

$$\left[-\frac{\hbar^2}{2m}\nabla^2 + V(r) + V_{\ell s}(r)\,\vec{L}\cdot\vec{S} - \varepsilon\right]\Psi(r) = 0$$

$$V_{\ell s}(r) \sim -\lambda\cdot\frac{1}{r}\cdot\frac{dV}{dr} \qquad \lambda > 0$$





Shell Model Potential

1 $$H = H_0 + V_{ls}(r)\vec{L}\cdot\vec{S}$$

$$H_0 = -\frac{\hbar^2}{2m}\nabla^2 + V(r)$$

$\vec{L}\cdot\vec{S}$ causes the reordering of the energy levels

- ***H_0 commutes with $L^2, L_z, S^2, S_z \Rightarrow$ energy levels have definite values of L^2, L_z, S^2, S_z***

- ***Wave func. for H_0 : $|l, m_l, s, m_s\rangle = \chi_s^{m_s} Y_l^{m_l}$, where l, s, m_l, m_s: good quantum numbers***

- ***H (with $\vec{L}\cdot\vec{S}$ term) commutes with L^2 and S^2 , but, does not commute with L_z and S_z***

- *Total angular momentum* , $\boxed{\vec{J} = \vec{L} + \vec{S}}$

- ***H commutes with J^2 and J_z***

$$J^2 = j(j+1)\hbar^2$$

$$J_z = m_j\hbar \qquad m_j = j, (j-1), \ldots -(j-1), -j$$

$$m_j = m_l + m_s$$

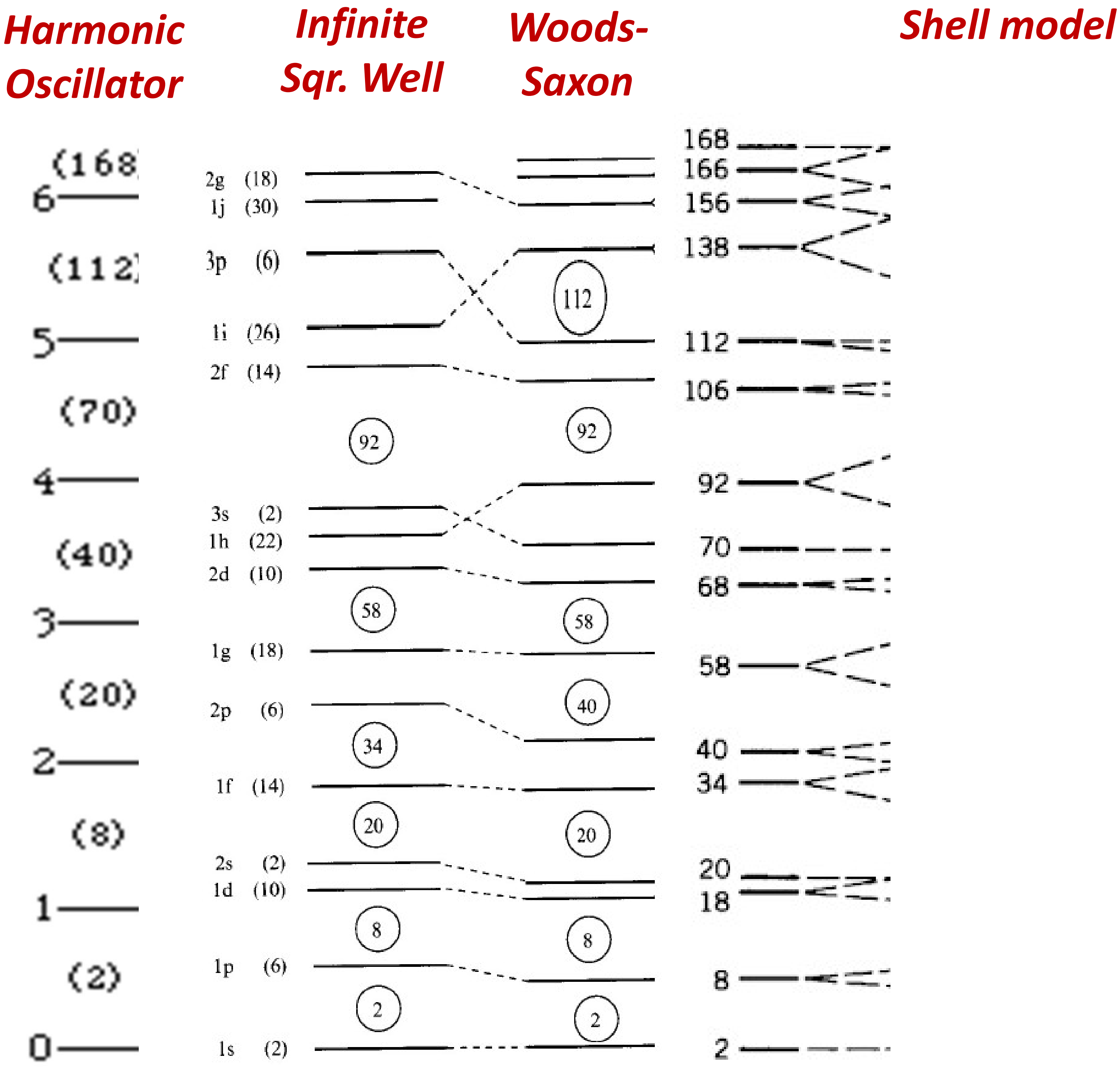
2

- ***j and $m_j (= m_l + m_s)$ can have definite values in the energy eigen functions***

- ***A state can now be defined in terms of quantum numbers of J^2 and J_z (j, m_j)***

- ***Wave func. For H : $|l, s, j, m_j\rangle$, where l, s, j, m_j : good quantum numbers***

Shell Model Potential



Shell Model Potential

$\vec{L}\cdot\vec{S}$ ***should be evaluated with the state defined by j* and** m_j **,** $|l, s, j, m_j\rangle$

Total angular momentum, $\vec{J} = \vec{L} + \vec{S}$ $\qquad j = l + s$

$$J^2 = (L+S)^2$$

$$\vec{L}\cdot\vec{S} = \frac{1}{2}\left(J^2 - L^2 - S^2\right)$$

$$\langle \vec{L}\cdot\vec{S}\rangle = \frac{1}{2}[j(j+1) - l(l+1) - s(s+1)]\hbar^2 \qquad (3)$$

$s = \frac{1}{2}$

$$l = 0 \Rightarrow j = l + \frac{1}{2}$$

$$l \neq 0 \Rightarrow j = l + \frac{1}{2}, l - \frac{1}{2}$$

$j = l + \frac{1}{2}$

$$\langle \vec{L}\cdot\vec{S}\rangle = \frac{1}{2}\left[\left(l+\frac{1}{2}\right)\left(l+\frac{3}{2}\right) - l(l+1) - \frac{1}{2}\left(\frac{1}{2}+1\right)\right]\hbar^2 = \frac{\hbar^2}{2}l \qquad (4)$$

$j = l - \frac{1}{2}$

$$\langle \vec{L}\cdot\vec{S}\rangle = \frac{1}{2}\left[\left(l-\frac{1}{2}\right)\left(l+\frac{1}{2}\right) - l(l+1) - \frac{1}{2}\left(\frac{1}{2}+1\right)\right]\hbar^2 = -\frac{\hbar^2}{2}(l+1) \qquad (5)$$

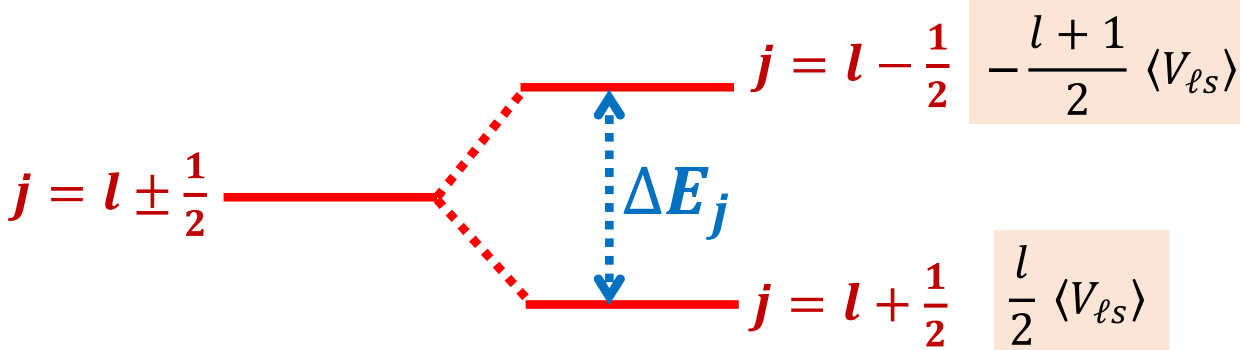
Shell Model Potential

$$j = l + \frac{1}{2}\ , \quad \langle \vec{L}\cdot\vec{S} \rangle = \frac{\hbar^2}{2} l \qquad \boxed{V(r) + \frac{l}{2} V_{\ell s}}$$ 4

$$j = l - \frac{1}{2}\ , \quad \langle \vec{L}\cdot\vec{S} \rangle = -\frac{\hbar^2}{2} (l+1) \qquad \boxed{V(r) - \frac{l+1}{2} V_{\ell s}}$$ 5

Splitting of energy levels corresponding to $j = l + \frac{1}{2}$ ***and*** $j = l - \frac{1}{2}$

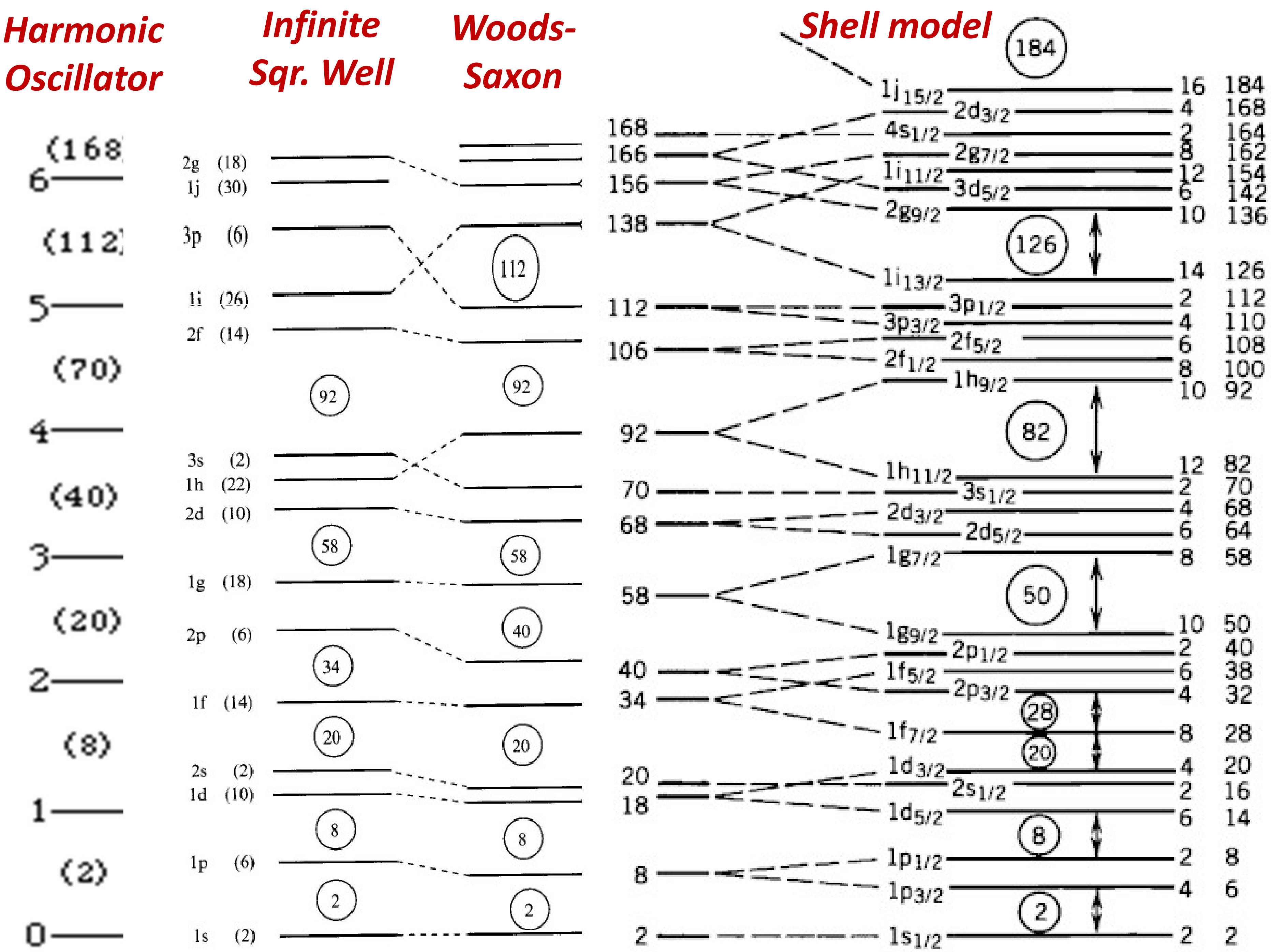
1 $$\boxed{H = H_0 + V_{ls}(r)\vec{L}\cdot\vec{S}}$$ $V_{ls}(r)$ *is* $-ve$



$$\Delta E_j = \frac{\hbar^2}{2} l + \frac{\hbar^2}{2}(l+1) = \frac{\hbar^2}{2}(2l+1)$$

larger $l \Rightarrow$ ***larger splitting***

Shell Model Potential



Extreme Single-Particle Shell model

1. Observed ground state spin and parity of nuclei

Spin of nucleus = total angular momentum, I^{π}

$$l = even \Rightarrow \pi = +ve$$

$$l = odd \Rightarrow \pi = -ve$$

Even-even nuclei

- *Nucleons are paired up in each quantum level*
- *Paired nucleons give zero spin and zero magnetic moment*
- *The paired nucleons form an inert core*
- *Predicted spin of e-e nuclei is zero*
- *All e-e nuclei in their ground state have*

$$I^{\pi} = 0^{+}$$

Extreme Single-Particle Shell model

1. Observed ground state spin and parity of nuclei

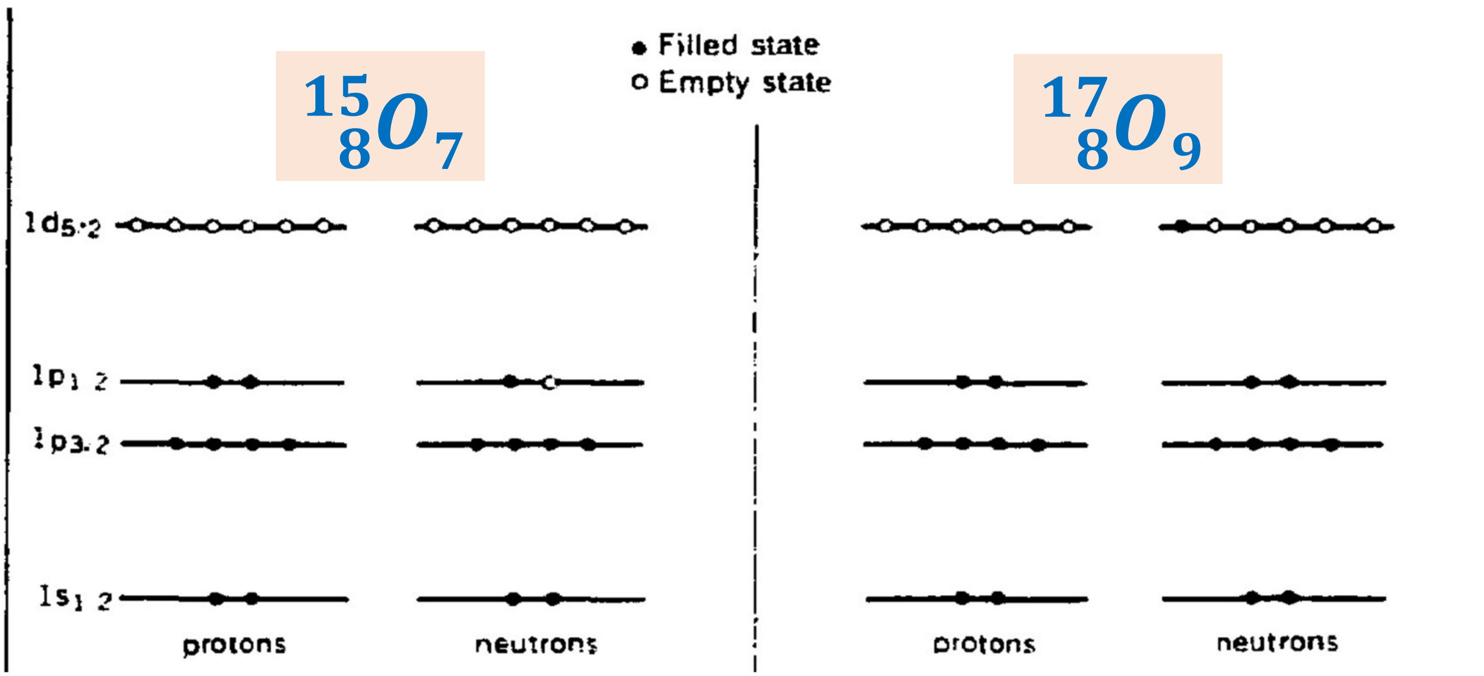
Spin of nucleus = total angular momentum, I^{π}

$l = even \Rightarrow \pi = +ve$

$l = odd \Rightarrow \pi = -ve$

Odd-A nuclei

- ***There is one neutron or proton that is unpaired***
- ***This last un-paired nucleon decides the properties of the nucleus***



Ground state $I^{\pi} = \frac{1}{2}^{-}$

Ground state $I^{\pi} = \frac{5}{2}^{+}$

Extreme Single-Particle Shell model

1. Ground state spin and parity of nuclei

Success of the extreme single-particle shell model

Every orbital has 2j+1 magnetic sub-states, completely filled orbitals have spin J=0, they do not contribute to the nuclear spin.

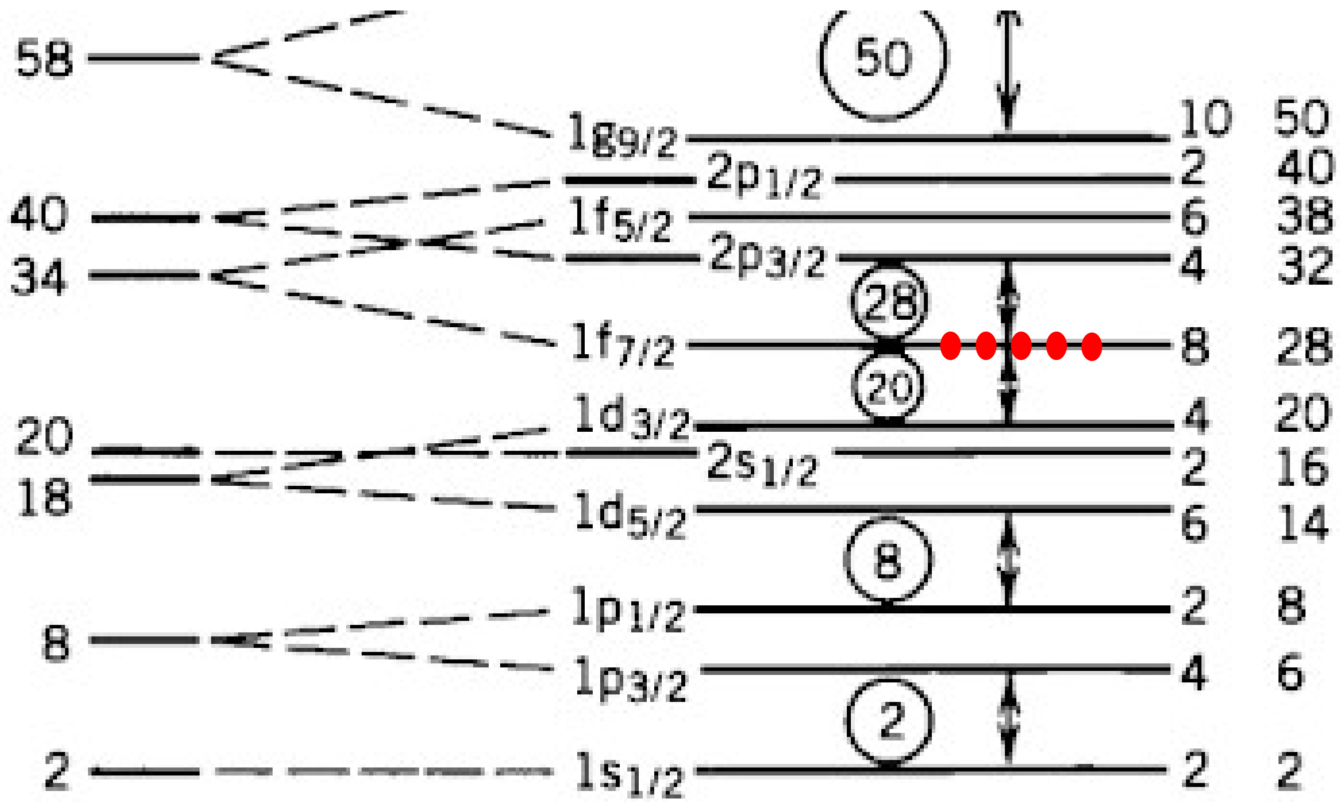
For a nucleus with one nucleon outside a completely occupied orbital the nuclear spin is given by the single nucleon.

$n\,\ell\,\boldsymbol{j} \rightarrow \boldsymbol{J}$

$(-)^{\ell} = \pi$

$^{53}_{25}Mn_{28}$

Observed $I^{\pi} = \frac{7}{2}^{-}$



Extreme Single-Particle Shell model

1. Ground state spin and parity of nuclei

Spin of nucleus = total angular momentum, I^{π}

$l = even \Rightarrow \pi = +ve$

$l = odd \Rightarrow \pi = -ve$

Odd-A nuclei

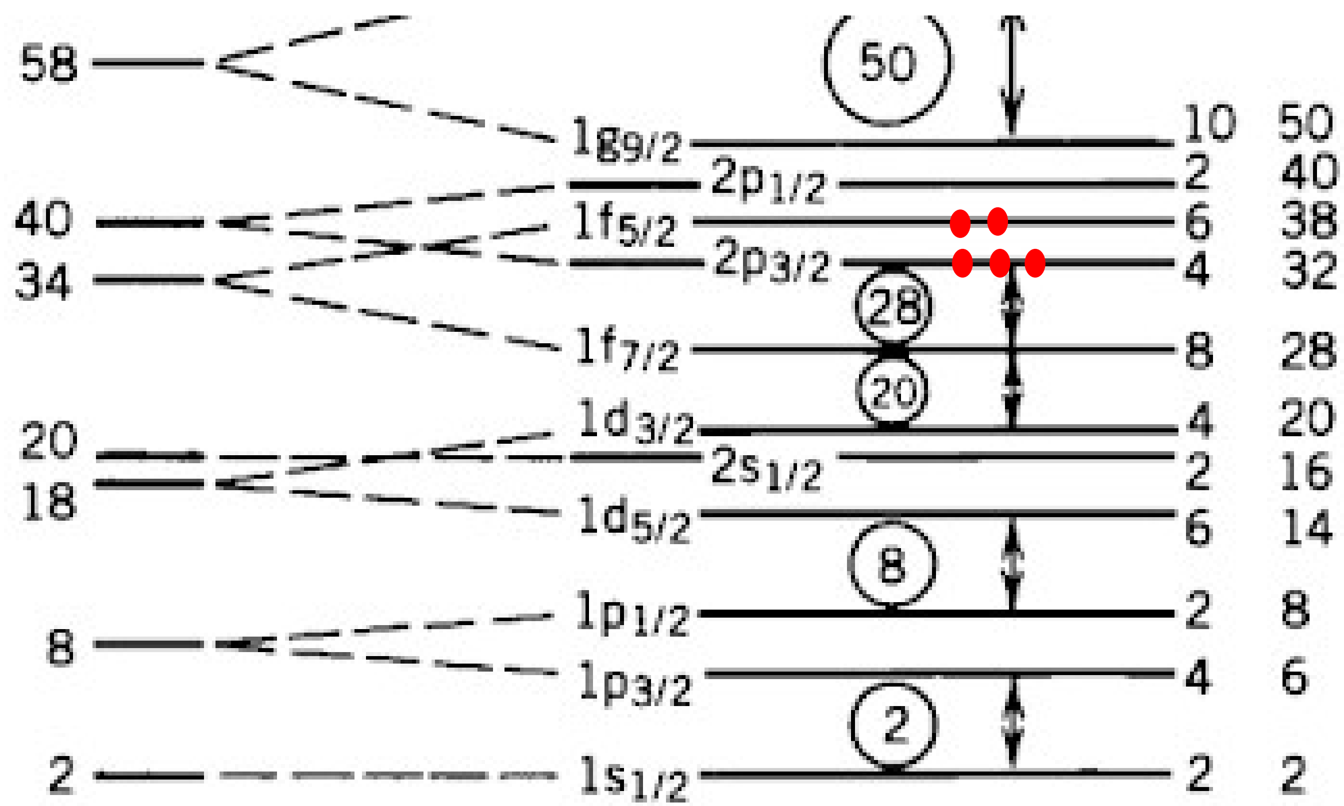
$^{61}_{28}Ni_{33}$

Expected $I^{\pi} = \frac{5}{2}^{-}$

Observed $I^{\pi} = \frac{3}{2}^{-}$

Pairing energy (amount by which the energy of the nucleus is reduced due to pairing) is larger if the pairing is made in a larger l state

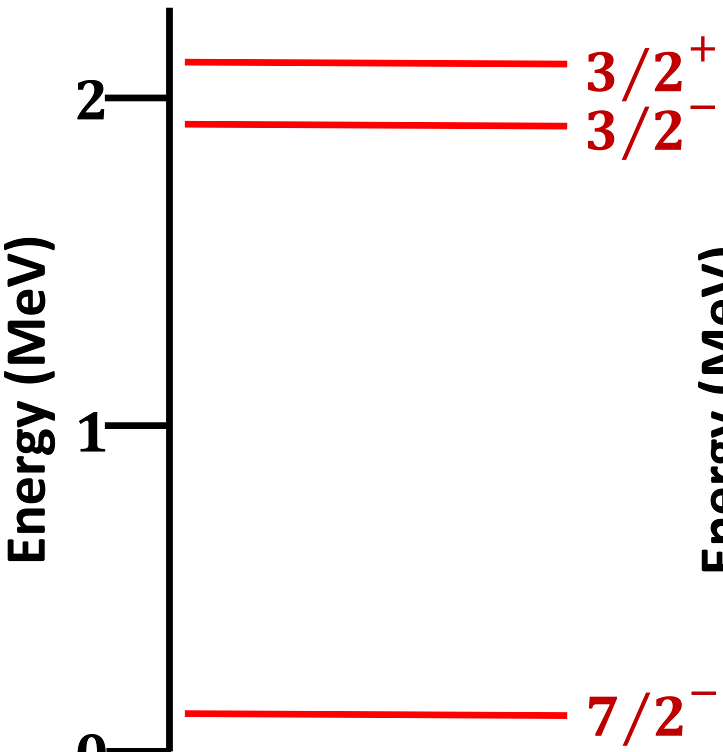
Amount of energy depends on l



Extreme Single-Particle Shell model

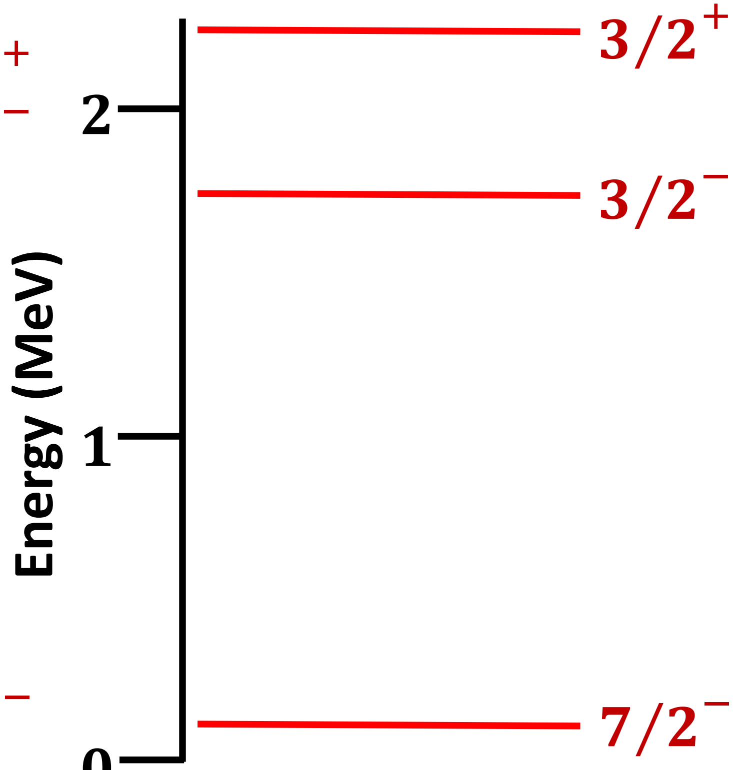
2. Excited states spin and parity of nuclei

$^{41}_{20}Ca_{21}$

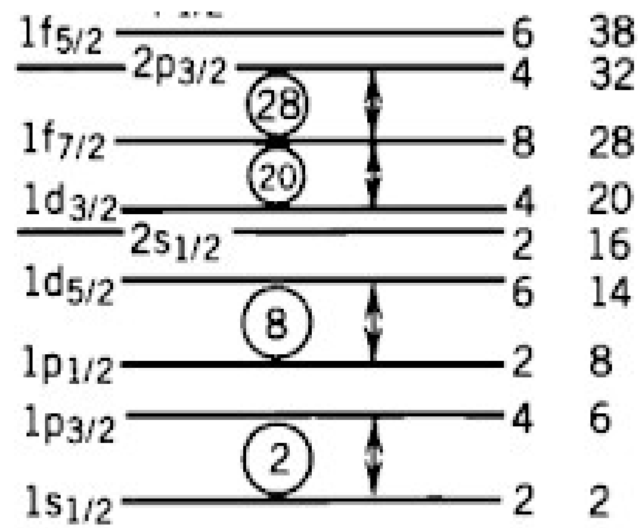


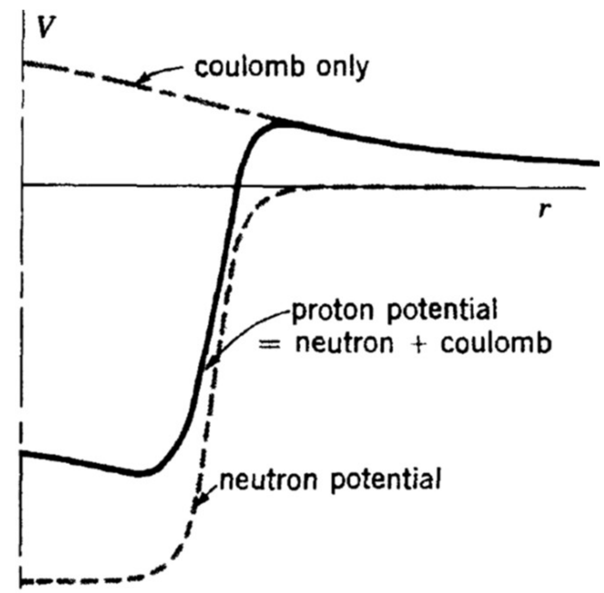
$Z = 20$
$N = 21$

$^{41}_{21}Sc_{20}$



$Z = 21$
$N = 20$

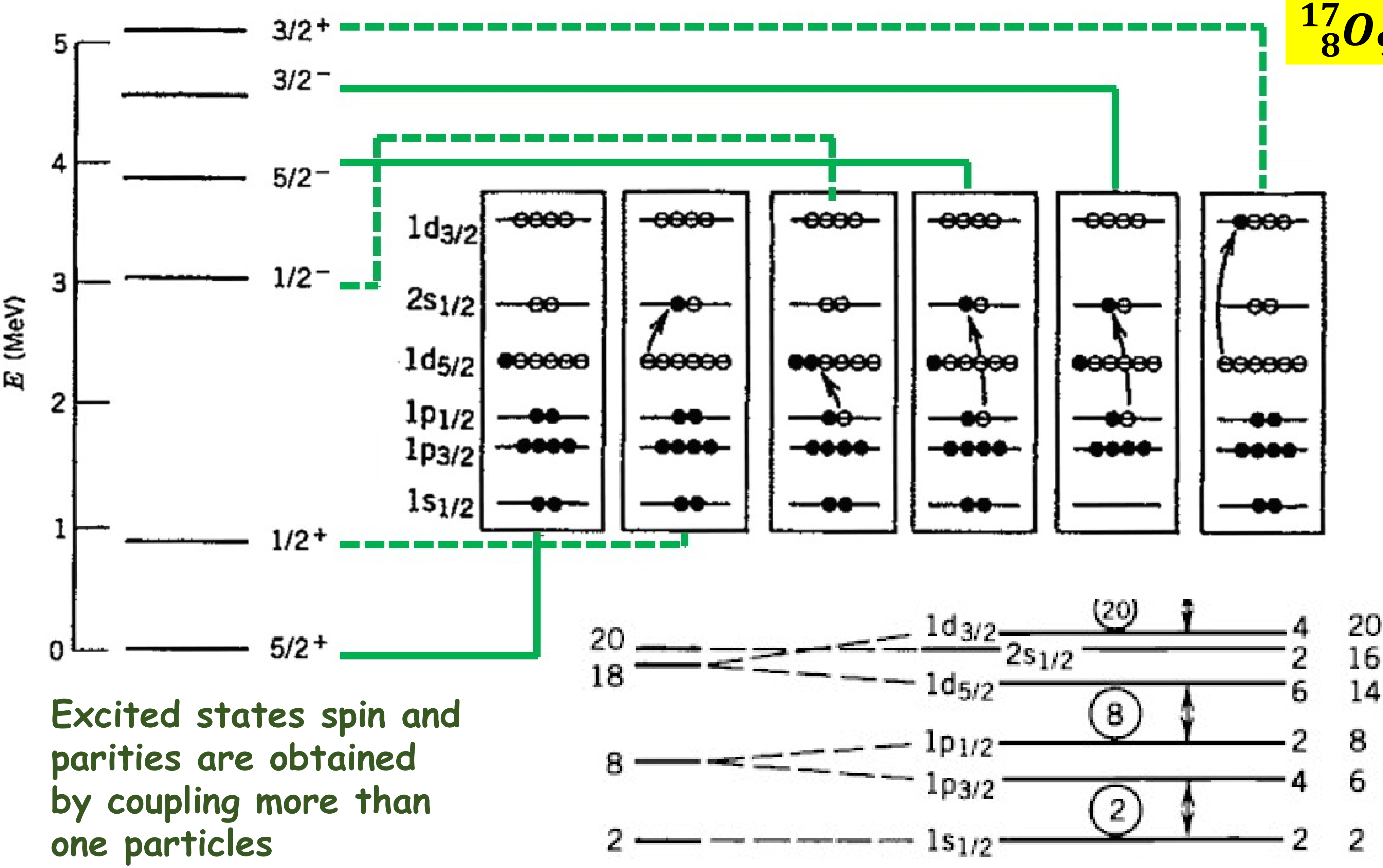




Extreme Single-Particle Shell model

2. Excited states spin and parity of nuclei

$^{17}_{8}O_{9}$



Excited states spin and parities are obtained by coupling more than one particles